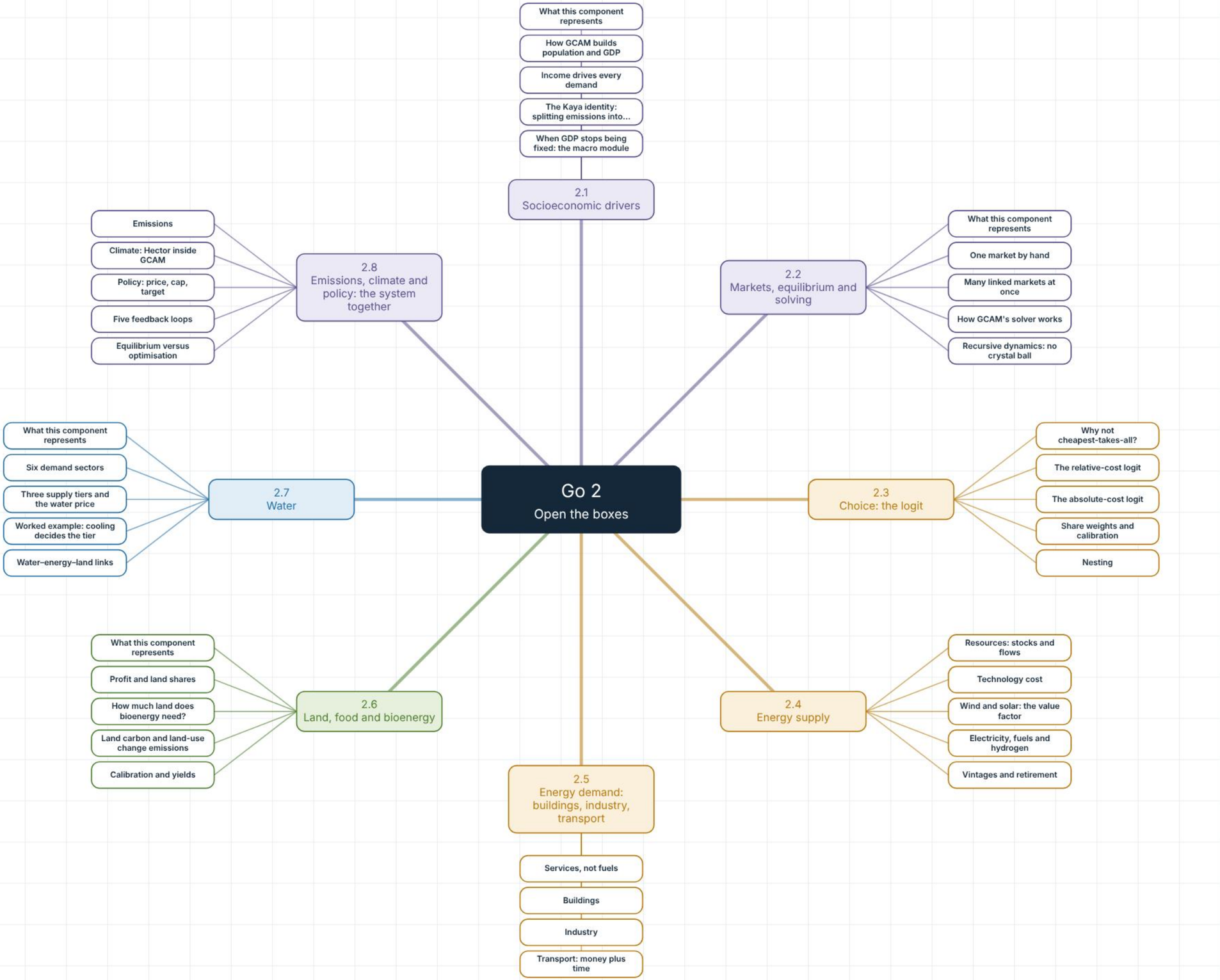
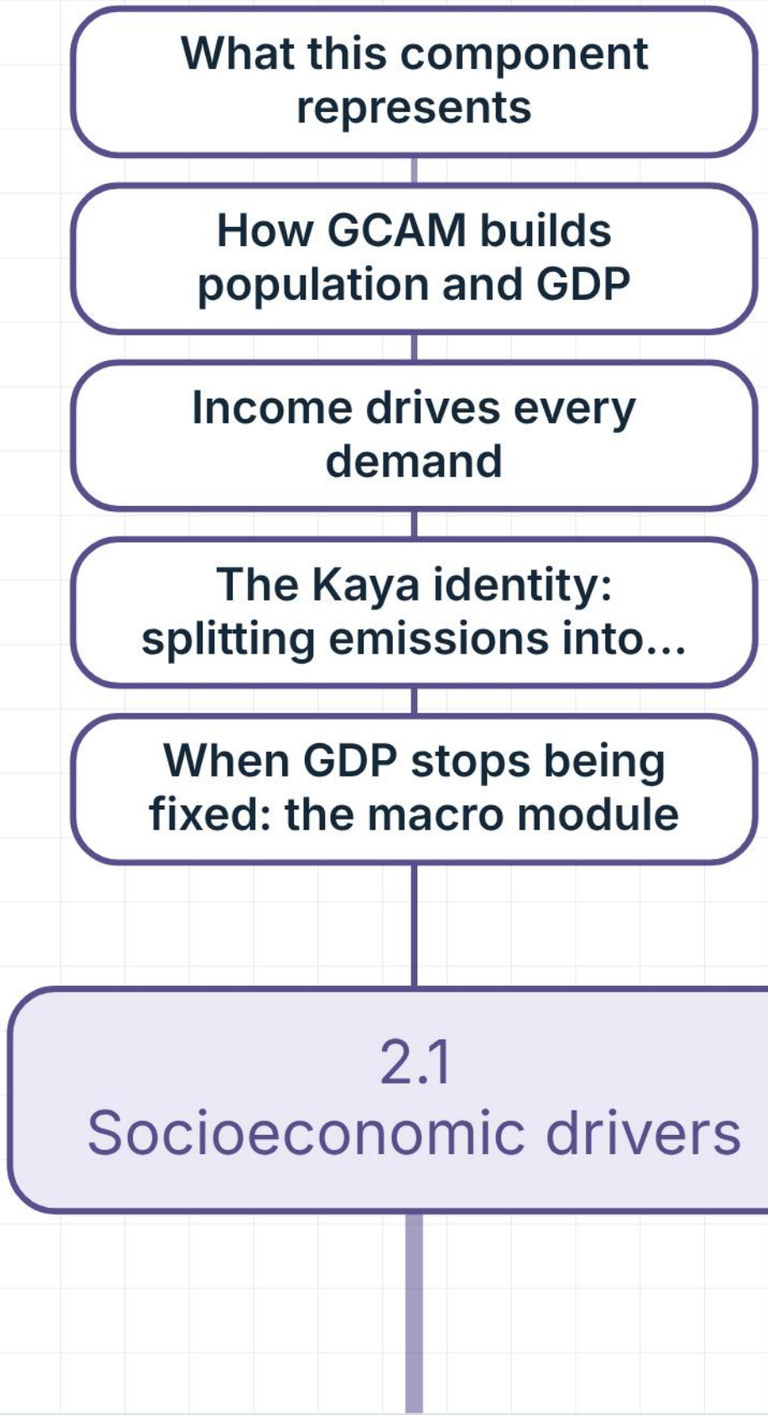


Go 2: Open the boxes

Open each GCAM component, see the rule it follows and the numbers it needs, and trace how it pushes on the others.

8 modules. Press Next to zoom into each one.



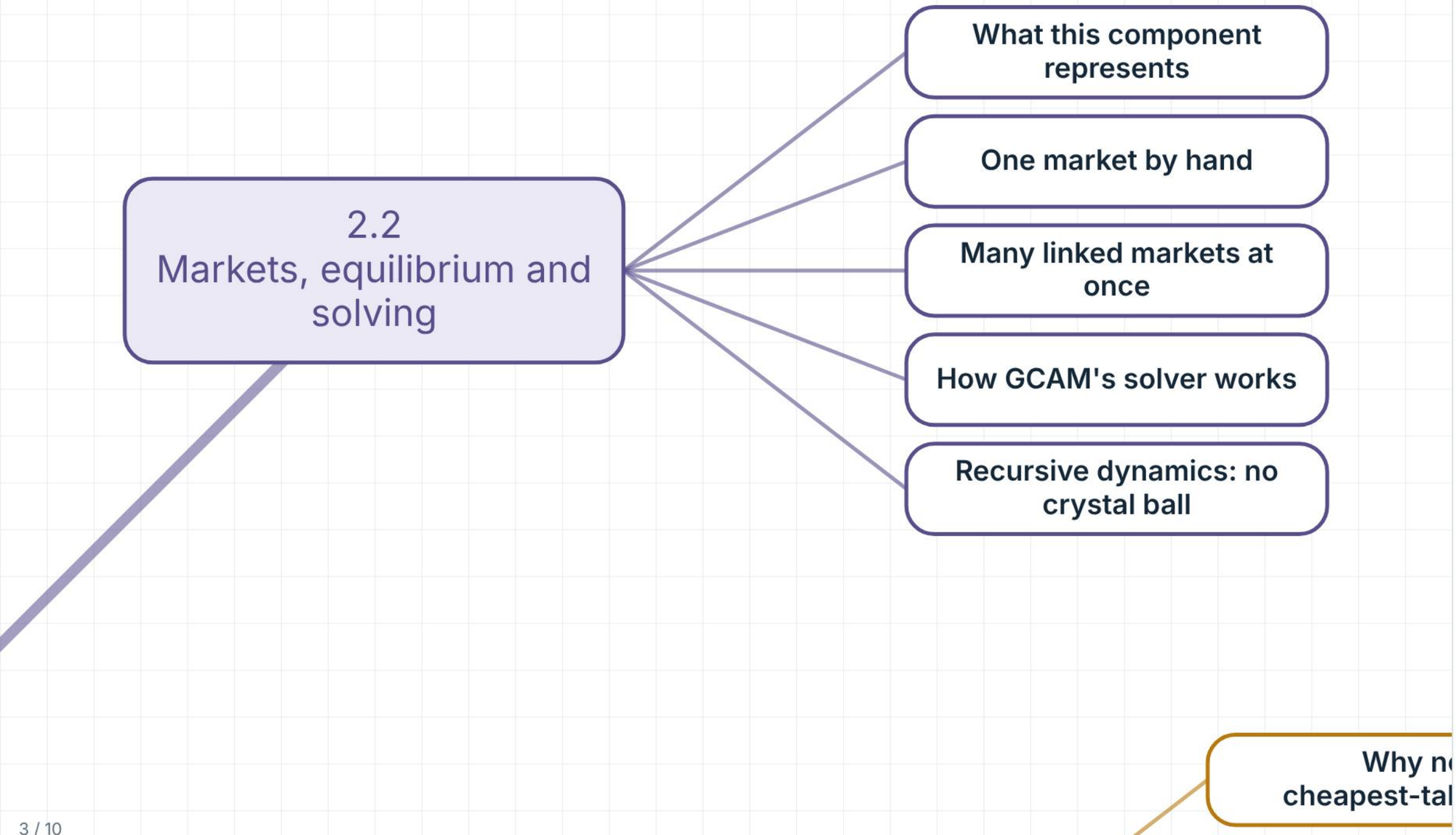
2.1 Socioeconomic drivers

How many people there are and how rich they become set the size of every demand in GCAM. Learn where population and GDP come from, and how the Kaya identity splits emissions into drivers.

Remember

- Population and labour productivity are inputs, usually taken from an SSP storyline
- GDP per person drives demand for buildings, transport, industry, food and water
- The Kaya factors multiply, so their ratios multiply: never add percentages
- GCAM runs 32 regions in 5-year steps from a 2021 base year to 2100

Open module 2.1



2.2 Markets, equilibrium and solving

GCAM is a giant auction: in each 5-year period it searches for the prices that make supply equal demand in every market at once. See how one market clears, then many linked markets.

Remember

- Equilibrium means supply equals demand in every market at the same time
- Excess demand F(p) is the gap the solver drives to zero
- Linked markets must be solved together; GCAM uses Broyden's method in log prices
- A solved period needs every market within tolerance; check the log

Open module 2.2

Recursive dynamics: no crystal ball

Why not cheapest-takes-all?

The relative-cost logit

2.3
Choice: the logit

The absolute-cost logit

Share weights and calibration

Nesting

Resources: stocks and flows

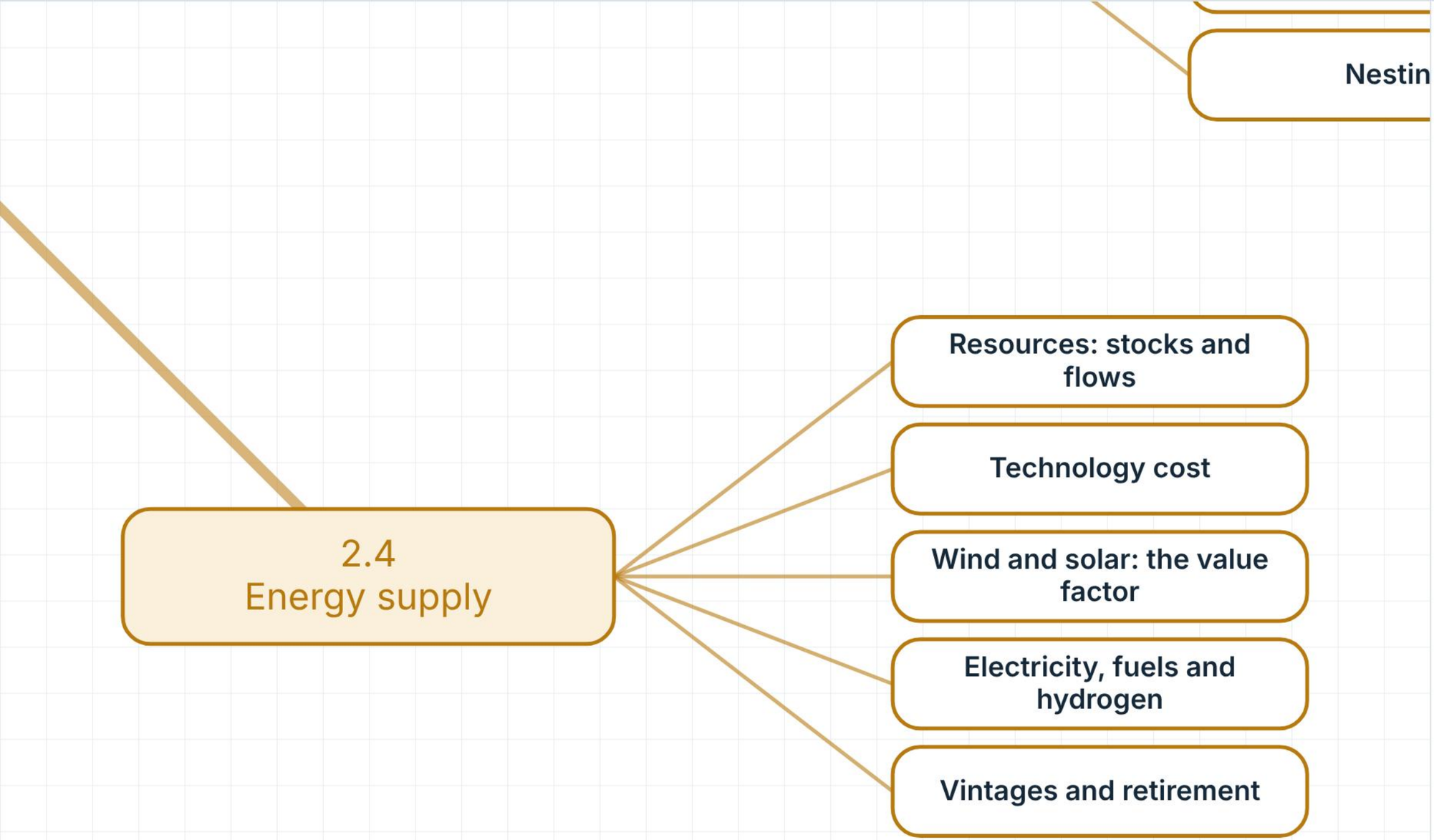
2.3 Choice: the logit

GCAM never gives the whole market to the cheapest option. A logit shares it out by relative cost, with share weights calibrated to history. Learn the formula, the exponent and nesting.

Remember

- The logit turns costs into smooth shares: cheaper wins more, never everything
- The exponent sets how sensitive shares are to cost differences
- Share weights calibrate the base year and phase in new options; future values are assumptions
- GCAM uses relative-cost and absolute-cost logits, nested from sector to technology

Open module 2.3



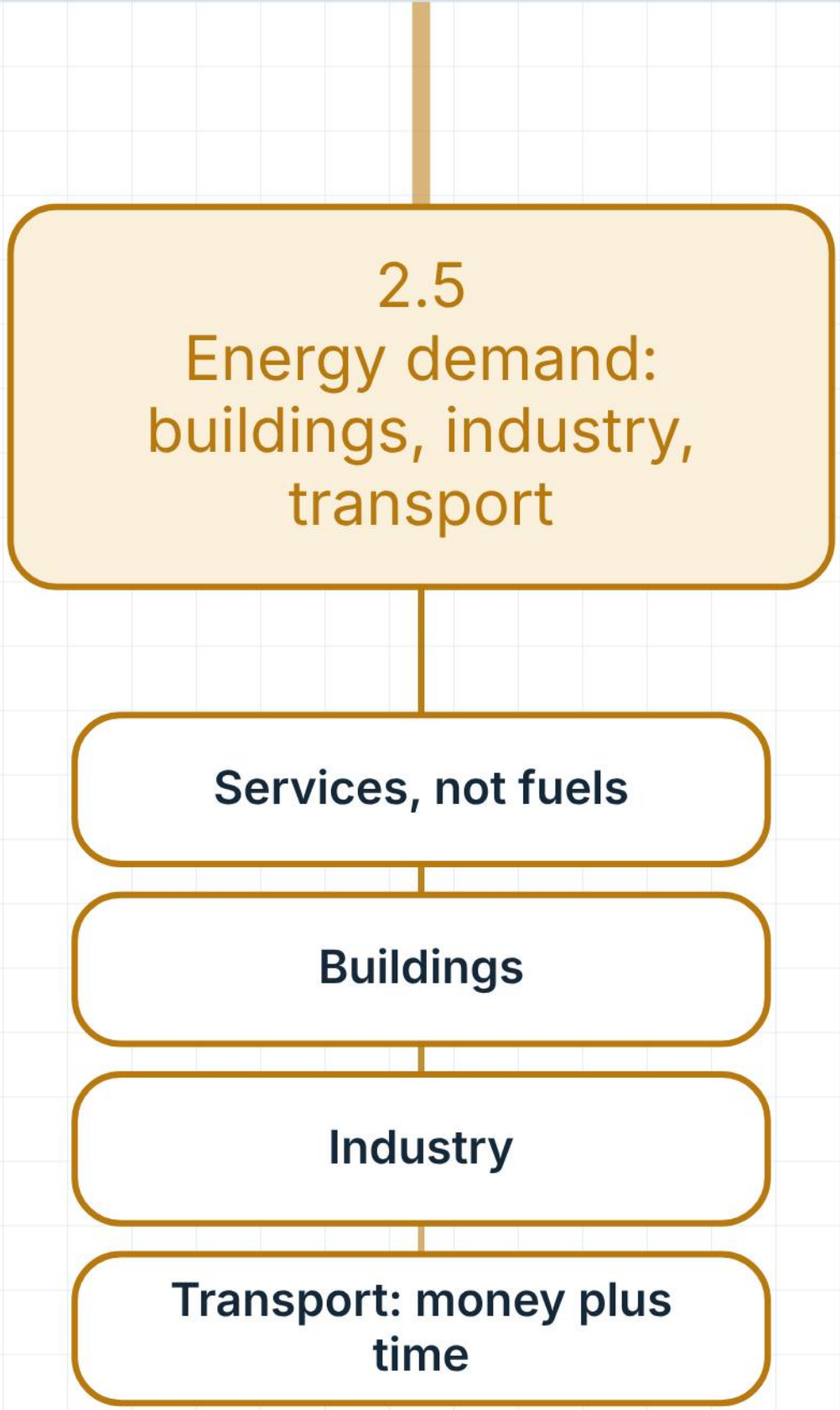
2.4 Energy supply

Where energy comes from in GCAM: graded fossil resources, renewable supply curves, power plants with levelised costs, and the value-factor rule that makes wind and solar dearer as they grow.

Remember

- Fossil fuels come from graded supply curves; renewables from flow supply curves
- Technology cost = non-energy cost + fuel cost + emissions cost
- A carbon price raises each option's cost in proportion to its emissions
- Wind and solar costs are divided by a value factor that falls with their share (v7.3+)

Open module 2.4



2.5 Energy demand: buildings, industry, transport

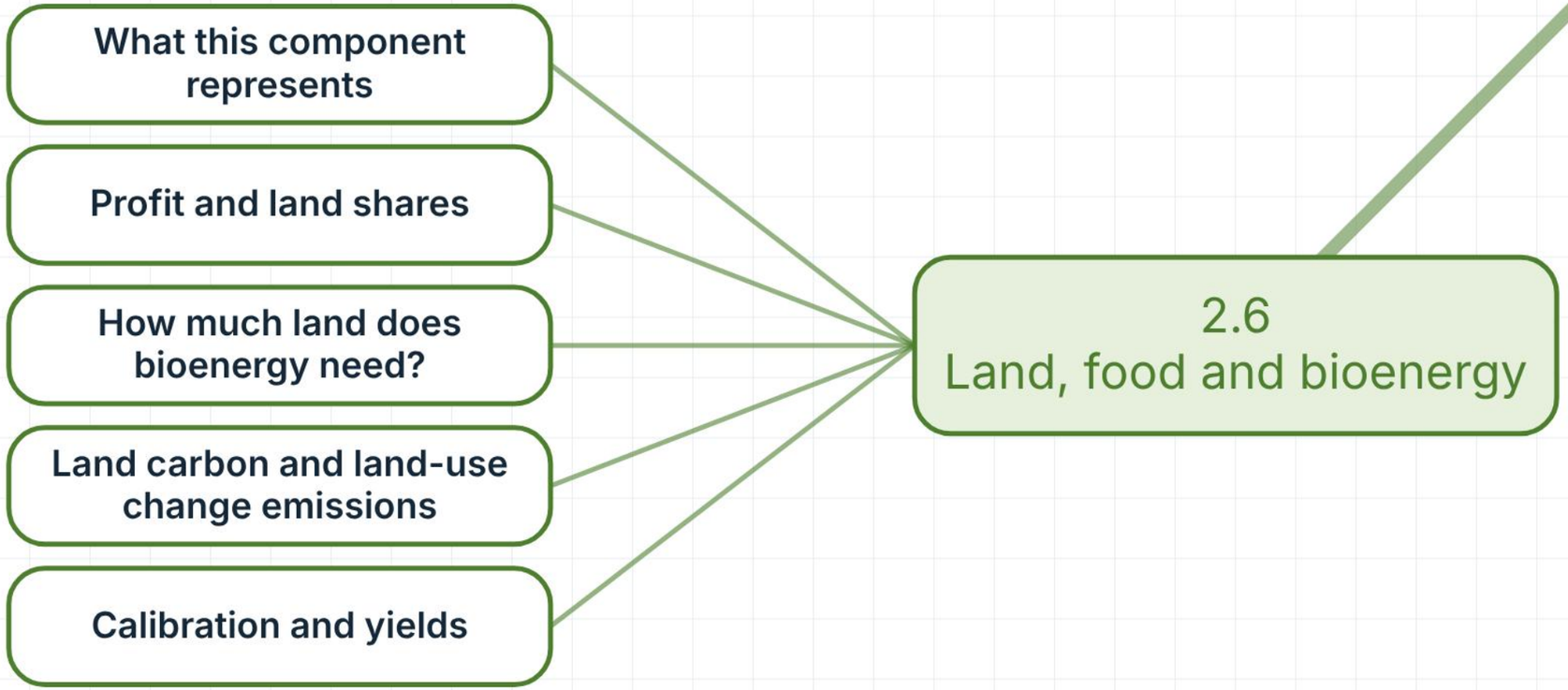
People want services, not energy: warm rooms, steel, passenger-kilometres. See how GCAM grows service demand with income and price, and how transport modes compete on money plus time.

Remember

- GCAM models energy services first, then the fuels that supply them
- Service demand rises with income (α) and falls with price (β)
- Buildings use floorspace and degree-days; industry is split into subsectors
- Passenger modes compete on money cost plus wage-based time cost

Open module 2.5

ergy–land links



2.6 Land, food and bioenergy

Land is shared among crops, pasture, forest and energy crops by a logit on profit per hectare. See why bioenergy demand raises food prices and how a carbon price can grow forests.

Remember

- Land is shared among uses by a logit on profit per hectare, inside a nest
- Bioenergy and food compete for cropland, so bioenergy demand can raise food prices
- Land needed = energy ÷ (yield × energy content): yield matters as much as the target
- Land-use change emits carbon fast and regrowth absorbs it slowly

Open module 2.6

Equilibrium versus optimisation

What this component represents

Six demand sectors

Three supply tiers and the water price

Worked example: cooling decides the tier

Water–energy–land links

2.7 Water

What this component represents

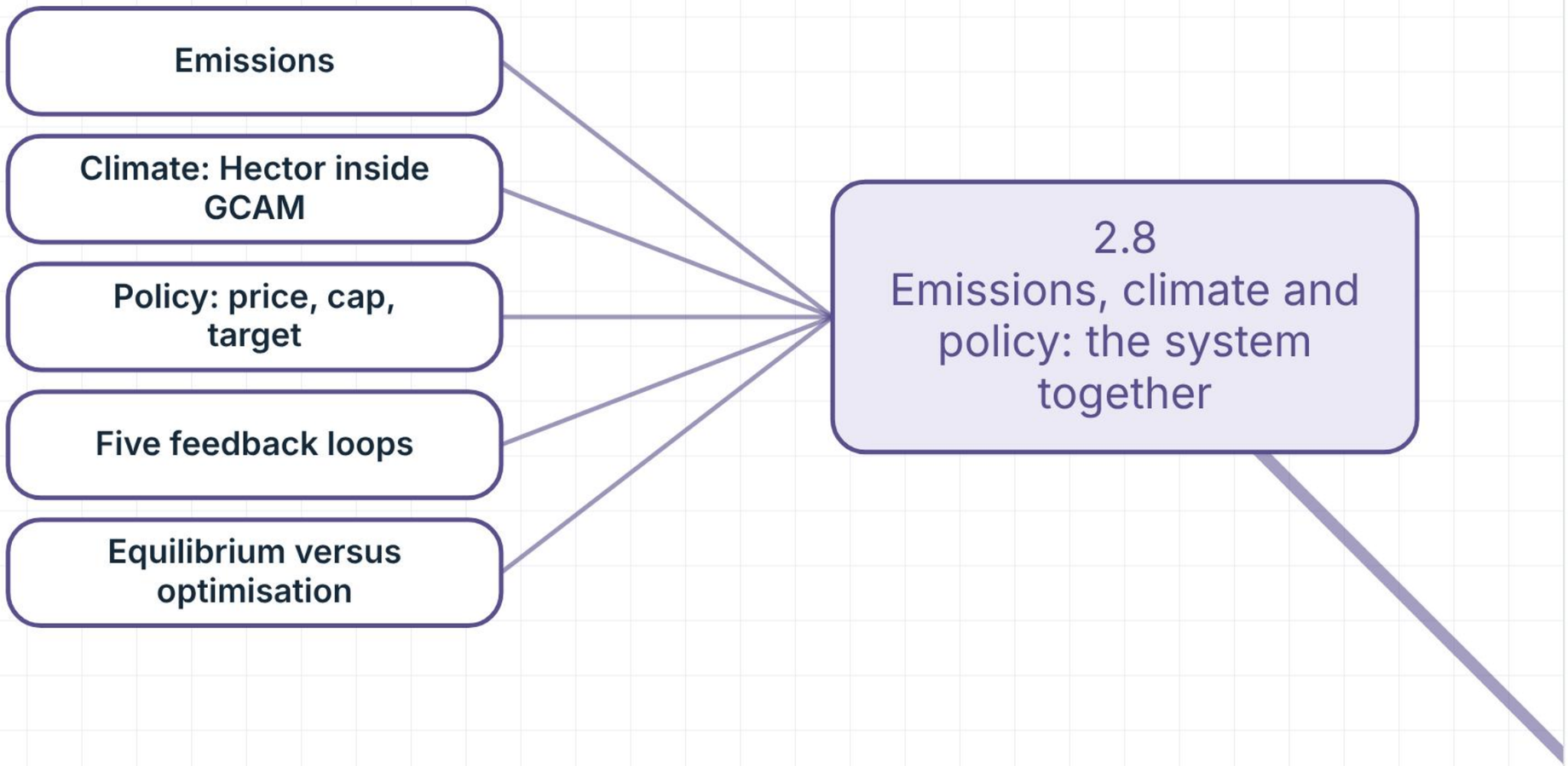
2.7 Water

GCAM balances water by river basin: cheap renewable water first, then groundwater, then desalination. See how withdrawals for farms, power and cities set the water price.

Remember

- Water is balanced by 235 basins with a basin price
- Six sectors withdraw: irrigation, electricity, manufacturing, primary energy, livestock, municipal
- Supply tiers: accessible renewable water, then groundwater, then desalination
- Cooling choice links electricity to water; desalination links water to electricity

Open module 2.7



2.8 Emissions, climate and policy: the system together

Put all the boxes back together. Follow emissions into Hector, see how GCAM implements carbon prices and caps, and trace five feedback loops that only an integrated model can show.

Remember

- Emissions from fuels, land and non-CO_2 gases go to Hector each period
- A carbon price fixes the price; a cap fixes emissions and GCAM finds the price
- Climate targets search for a price path and take much longer to run
- One policy moves energy, land, water and emissions together through feedback loops

Open module 2.8